EXPERIMENT-23 Family Tree

Aim

To implement a family tree in Prolog and define mother, father, grandfather, grandmother, sister, and brother relations.

Objectives

- To represent family relationships using facts.
- To define relationships using rules.
- To perform logical queries on the family tree.

Algorithm

1. Define male and female persons.
2. Define parent relationships.
3. Define mother and father using gender and parent.
4. Define grandparents using parent relationships.
5. Define siblings based on common parents.
6. Define sister and brother based on sibling and gender.

Flowchart

```
      START
        |
        v
  Define male/female facts
        |
        v
   Define parent facts
        |
        v
```

```
    Define family relations
            |
            v
      Enter a query
            |
            v
    Prolog searches facts
            |
            v
      Display result
            |
            v
          STOP
```

Program

```
% Family Tree

female(pam).
female(liz).
female(ann).
female(pat).

male(tom).
male(bob).
male(jim).
```

```
% Parent relationships
parent(pam, ann).
parent(tom, ann).

parent(pam, liz).
parent(tom, liz).

parent(ann, pat).
parent(bob, pat).

parent(ann, jim).
parent(bob, jim).

% Mother
mother(Mother, Child) :-
    female(Mother),
    parent(Mother, Child).

% Father
father(Father, Child) :-
    male(Father),
    parent(Father, Child).

% Grandfather
grandfather(Grandfather, Child) :-
```

```
    male(Grandfather),
    parent(Grandfather, Parent),
    parent(Parent, Child).

% Grandmother
grandmother(Grandmother, Child) :-
    female(Grandmother),
    parent(Grandmother, Parent),
    parent(Parent, Child).

% Sibling
sibling(X, Y) :-
    parent(P, X),
    parent(P, Y),
    X \= Y.

% Sister
sister(Sister, Person) :-
    female(Sister),
    sibling(Sister, Person).

% Brother
brother(Brother, Person) :-
    male(Brother),
    sibling(Brother, Person).
```

Queries

?- mother(pam, ann).

Output:

true.

?- father(tom, liz).

Output:

true.

?- grandfather(tom, pat).

Output:

true.

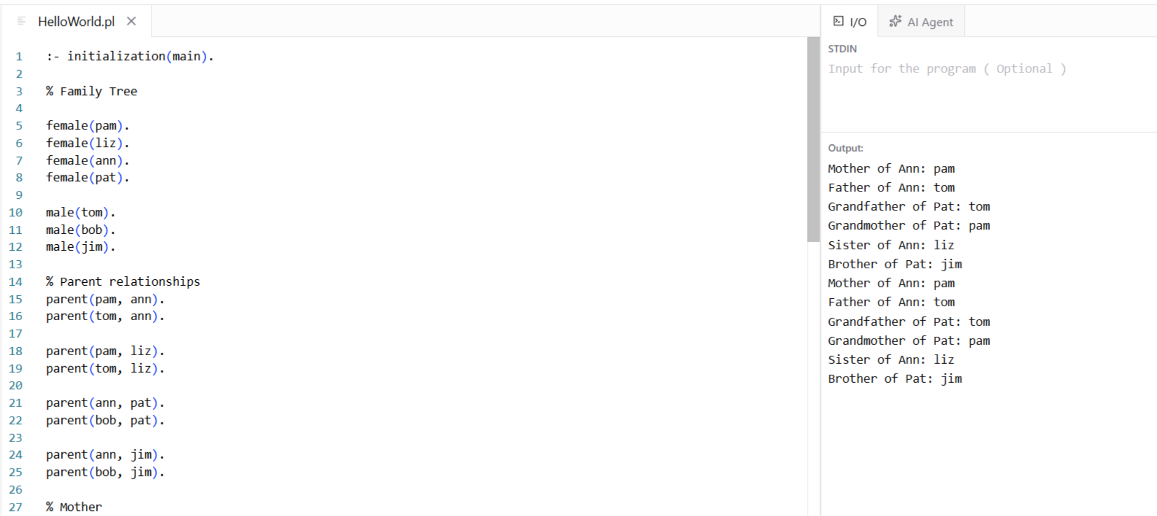
?- grandmother(pam, jim).

Output:

true.

?- sister(liz, ann).

Output:



Result

The family tree and required relationships were successfully implemented.

Conclusion

Prolog is suitable for representing family relationships and answering relationship-based queries.